



Tribhuvan University

Faculty of Humanities and Social Sciences

**A PROJECT REPORT ON
“ONLINE STUDENT ATTENDANCE MANAGEMENT SYSTEM”**

**Submitted to
Department of Computer Application
VS INTERNATIONAL COLLEGE**

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted by
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Under the Supervision of
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Tribhuvan University

Faculty of Humanities and Social Sciences

VS INTERNATIONAL COLLEGE

SUPERVISOR RECOMMENDATION

I hereby recommend that this project prepared under my supervision by **Srijana Neupane and Sunita Tamang** entitled “**ONLINE STUDENT ATTENDANCE MANAGEMENT SYSTEM**” in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

.....

Mr. Sheejan Adhikari

Supervisor



Tribhuvan University

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LETTER OF APPROVAL

This is to certify that this project prepared by **SRIJANA NEUPANE** and **SUNITA TAMANG** entitled “**ONLINE STUDENT ATTENDANCE MANAGEMENT SYSTEM**” in partial fulfillment of the requirements for the degree of Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

..... Mr. Sheejan Adhikari Supervisor	 Mr. Santosh Pokharel Coordinator
..... Mr. Rajkumar Acharya Principal	 External Examiner

ABSTRACT

The Online Student Attendance Management System is a web-based application developed to automate and improve the traditional process of recording and managing student attendance in educational institutions. In many colleges and schools, attendance is still maintained manually using paper registers or spreadsheets, which is time-consuming, inefficient, and susceptible to errors, data loss, and manipulation.

The proposed system allows administrators to centrally manage student/teacher records, courses, through a unified platform. Faculty members (Teachers) can log in securely and mark attendance online to record daily attendance for their respective classes with ease, reducing the time spent on administrative tasks. Students are given access to view their attendance in real time, which promotes transparency and encourages regular attendance. The system maintains accurate and consistent data, enabling to monitor attendance patterns effectively, quick retrieval of records, and reduced chances of manipulation or loss of information. Features such as role-based access control, user authentication, and secure database management ensure data integrity and confidentiality. The system minimizes redundancy, eliminates manual calculations, and reduces the chances of human error.

The Online Student Attendance Management System is developed using modern web technologies, including HTML, CSS for the frontend interface, PHP for server-side processing, and MySQL for backend database management. This system is designed to be reliable, secure and user-friendly.

Keywords: *Online Student Attendance, Attendance Management System, Web-Based Application, Student Attendance view, Faculty Management, Role-Based Access, User Authentication, Secure Database, Data Integrity, Digital Attendance, Automation, PHP, MySQL, HTML, CSS, JavaScript, Educational Institutions.*

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We acknowledge our institution for providing the necessary resources, laboratory facilities, and a supportive environment, which were essential for carrying out this project effectively.

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Srijana Neupane

Sunita Tamang

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LIST OF ABBREVIATIONS

CASE	–	Computer Aided Software Engineering
CSS	–	Cascading Stylesheet
DFD	–	Data Flow Diagram
ER	–	Entity Relationship
HTML	–	Hypertext Markup Language
OSAMS	–	Online Student Attendance Management System
PHP	–	Hypertext Preprocessor
RBA	–	Role Based Access
UI	–	User Interface
VS Code	–	Visual Studio Code

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CHAPTER I: INTRODUCTION

1.1 Introduction:

The Online Student Attendance Management System is a digital platform designed to replace traditional manual methods of recording and managing student attendance with a modern, efficient, and user-friendly solution that supports administrators, teachers, and students. Manual attendance, which often relies on paper registers or spreadsheets, is time-consuming, prone to errors, and difficult to maintain in large classes, leading to misplaced records and inconsistencies. By contrast, an online system automates attendance processes, reduces repetitive administrative work, and ensures that data is stored centrally and securely, minimizing errors and saving valuable time for teaching and management tasks. The system is designed with role-based access control, meaning that each user can access only the features relevant to their role. Administrators manage records centrally, teachers handle attendance for their classes, and students can view only their own records. This separation of responsibilities enhances security and ensures that sensitive data is protected while keeping the system easy to navigate for all users. The system allows teachers to register, log in, and log out securely, and then mark attendance for their respective classes electronically. Students interact with the system through their own accounts, allowing them to register, log in, log out, and view their attendance records in real time. In addition, the system provides real-time updates, enabling administrators and teachers to monitor attendance trends instantly and take timely action when necessary. The centralized database ensures that records are protected from loss, damage, or unauthorized modification.

Furthermore, the system improves transparency and communication within the institution. Students can regularly monitor their attendance status, which encourages responsibility and punctuality. Administrators can efficiently manage large volumes of student data without relying on physical storage space. Overall, the Online Student Attendance Management System enhances efficiency, security, and reliability in managing attendance, making it a valuable solution for modern educational institutions.

1.2 Problem Statement:

In most of the schools and colleges, attendance is still recorded manually in papers which are causing issues like: Time-consuming, prone to errors, and difficulty in managing, storing, and organizing large volumes of physical paper records; Inability in accessing real-time attendance information; Susceptible to unauthorized changes, making the records unreliable; No real-time access to the attendance records and Vulnerable to loss, damage, or manipulation, making it difficult to maintain accurate and secure records. Furthermore, manual attendance systems increase administrative workload and reduce overall efficiency in the academic environment.

1.3 Objectives:

The proposed Online School Attendance Management System aims to solve these problems by achieving the following objectives:

1.3.1 Specific Objectives:

- To implement role-based access that lets students and teachers register, teachers mark/view/ attendance, admins manage/authorize users and view/generate attendance records, and students log in/out and view their own attendance, ensuring secure and controlled access for all users.

1.3.2 General Objectives:

- To replace physical logbooks with a secure digital database, ensuring that attendance data is permanent, searchable, and protected from physical damage or loss.
- To provide a user-friendly interface that allows teachers to record daily attendance in under a minute, minimizing the distraction from actual teaching.

1.4 Scope and Limitation:

1.4.1 Scope:

- Saves a considerable amount of time and effort by automating the process of recording attendance.
- Minimizes human errors that commonly occur in manual entry.
- Authorized personnel can conveniently access and monitor attendance in real time.
- Attendance data is stored securely in digital databases, making it more reliable, protected from loss or tampering.

1.4.2 Limitation:

- The system is designed only for academic purposes, so it may not fully support corporate or industrial attendance requirements.
- The system neither provides a mobile application nor supports accessibility features for differently-abled individuals.
- The system currently includes core academic functions, such as marking attendance, viewing records, and providing authorization to users and may not support advanced features like biometric integration, automated notifications, or analytics dashboards.
- The system has been tested for small to medium datasets during the project duration (e.g. a single semester or a few classes), and may not efficiently handle very large datasets, such as universities with thousands of students, or support long-term scalability beyond the demonstration period.

1.5 Report Organization:

This report is structured into four main chapters that detail the development lifecycle of the project:

Chapter 1: Introduces the Project, its Objectives, Scope, and Limitations.

Chapter 2: Provides Background Study and Literature Review which establishes the theoretical foundation and explores existing research to contextualize the project's objectives.

Chapter 3: Focuses on System Analysis and Design, offering a deep dive into functional and non-functional requirements, feasibility studies, and comprehensive modeling including ER-Diagrams and DFDs to map out the system's architecture and interface.

Chapter 4: Provides the Implementation and Testing details, which specifies the CASE tools and programming languages used, describes the core modules, and outlines the unit and system testing protocols used for validation.

Chapter 5: Presents the Conclusion and Future Recommendations, summarizing the lessons learned and the overall project outcome while suggesting potential avenues for future development.

Chapter II: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study:

Attendance management is a critical function in schools and colleges, yet the traditional method of tracking student attendance—typically involving a physical register and verbal roll call—is increasingly inefficient and error-prone, consuming valuable lecture time and being susceptible to proxy attendance and data inaccuracies [1]. Manual attendance tracking has been shown to be time-consuming, often taking significant instructional time and requiring extensive administrative effort [2]. With the advancement of computer technology and the internet, manual attendance systems are increasingly replaced by online systems that leverage digital platforms and centralized databases to improve reliability, accessibility, and transparency [3]. An Online Attendance Management System enables teachers to record attendance digitally using computers or mobile devices, with secure storage in a centralized database that can be accessed on demand, ensuring consistency and reducing paperwork and human error [1], [3]. These systems can automatically generate attendance summaries—such as daily, monthly, and subject-wise reports—significantly reducing administrative burden, while administrators can monitor attendance records in real time, and students can view their attendance status, promoting accountability and discipline [1], [3]. Modern literature highlights the technological shift from barcode and RFID methods toward more advanced biometric techniques, especially face recognition, which can eliminate fraudulent attendance entries and improve accuracy; face recognition-based systems have been widely studied and implemented in educational attendance applications with promising results [4], [5]. Cloud-integrated attendance systems further enhance scalability by enabling seamless real-time data synchronization across institutional departments and facilitating remote access to attendance data [6]. Many systems also integrate mobile notifications and reporting dashboards, which alert teachers and administrators to low attendance rates or irregular patterns automatically. As educational institutions transition to hybrid and digital learning models, the need for a centralized, real-time attendance tracking system—especially one based on biometric authentication—has become increasingly important to support efficient academic management and reduce the limitations of traditional attendance methods.

2.2 Literature Review:

The conceptual framework of an Online Student Attendance Management System is built upon the interaction of three primary stakeholders: administrators, teachers, and students. Each stakeholder plays a distinct role within the system, contributing to its functionality and overall effectiveness. At the conceptual level, the system operates as a centralized web-based platform designed to automate attendance recording and management. Unlike manual systems, where data is scattered across registers or spreadsheets, the conceptual model of an online system integrates user authentication, database storage, and role-based access control into a single platform. This structure ensures that data flows efficiently between users while maintaining security and consistency. The administrator functions as the system controller. Conceptually, administrators manage user accounts, including adding teachers and students, updating records, and overseeing overall system operations. This centralized control ensures uniformity and reduces redundancy. Centralized databases enhance organizational efficiency by reducing duplication and improving data accessibility. In the attendance system, administrators also monitor attendance records to ensure compliance with institutional policies [8]. Teachers represent the operational users of the system. Conceptually, teachers interact with the system by registering, logging in, and marking attendance for their respective classes [9]. The system provides an interface where teachers can select courses and record student presence electronically. Digital marking significantly reduces time spent on administrative tasks and minimizes human error. Teachers can also view attendance records, enabling them to identify irregular attendance patterns and provide guidance to students when necessary [10]. Students are the end-users who benefit from transparency and accessibility. Conceptually, students register, log in, log out, and view their attendance records in real time. This visibility promotes accountability and encourages consistent attendance [11]. The system's architecture typically includes a frontend interface (developed using HTML, CSS, and JavaScript), a backend processing layer (using PHP or similar server-side technologies), and a relational database such as MySQL for data storage [12]. This layered structure supports efficient data processing and secure storage. The conceptual model follows a client-server architecture, where users access the system through web browsers, and the server handles authentication, data processing, and database transactions. Another important conceptual element is role-based access control (RBAC). RBAC ensures that administrators, teachers, and students have access only to the

features relevant to their responsibilities. Access control mechanisms are essential for maintaining data integrity and preventing unauthorized modifications. In the attendance system, administrators manage records, teachers mark attendance, and students only view their data, ensuring a structured and secure workflow [10]. The system also supports scalability within defined academic contexts. While designed primarily for small to medium-sized institutions, the conceptual framework allows future enhancements, such as integration with learning management systems (LMS), biometric authentication, or mobile applications [13]. Technological integration improves system reliability and user trust, further strengthening attendance management practices. Moreover, the conceptual framework emphasizes real-time data availability. Unlike manual systems, where updates may be delayed, digital systems instantly reflect attendance entries in the database. This real-time capability improves transparency and reduces disputes regarding attendance status. The timely data access supports proactive academic monitoring and intervention. Additionally, modern conceptual models incorporate analytics dashboards that help administrators and instructors visualize attendance trends over time, enabling data-driven decision making and early intervention for at-risk students [8],[11]. Advanced modules can generate automated alerts for poor attendance and integrate notifications via email or SMS, enhancing communication efficacy between stakeholders [9], [14]. Security considerations, such as encryption of sensitive data both at rest and in transit, are also integral components of the framework to protect against unauthorized access and ensure compliance with data protection standards [12]. By adopting responsive design principles, the system can be accessed seamlessly across desktop and mobile devices, supporting flexibility for users in diverse learning environments [12]. Recent extensions of the conceptual model propose integration with biometric and facial recognition systems to further mitigate proxy attendance and enhance authentication accuracy [6]. Ultimately, the conceptual framework demonstrates that an online attendance system delivers not only operational efficiency but also strategic insights that can improve teaching effectiveness and student engagement in modern academic administration [8].

CHAPTER III: SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis:

The system analysis of the system was done by conducting requirement analysis, feasibility analysis, data modeling, and process modeling as follows:

3.1.1 Requirement Analysis:

3.1.1.1 Functional Requirement:

The requirement that has been used in the project as the functional requirements generally includes the function such as inputs, the processing and the final output. The functional requirements in the project are mentioned below:

- Registration and Login/Logout.
- Student and Teacher Management & Report Generation by admin.
- Attendance Marking by teacher.
- Attendance Viewing by students, teachers and admin.

Below is the Use case diagram to show the functional requirement of the system as well as the workflow of the system.



Figure 3.1: Use case diagram

3.1.1.2 Non-Functional Requirement:

The non-functional requirement specifies how the system works. The non-functional requirements included in the project are:

- **Availability:** The system is available online. Being a web-based platform, the system can be accessed from any modern browser such as Chrome, Firefox, or Edge.
- **Security:** This system is secure and the users' information won't be available for others for users' privacy. The user password is in encrypted format in the database.
- **Performance:** This system is designed for smooth performance with optimization and good response.

3.1.2 Feasibility Analysis:

Following feasibilities were studied before building the system to see if the system could be built with exact requirements in required time.

3.1.2.1 Technical feasibility:

In order to design this system, it uses off-the-shelf and existing technologies, software and hardware so there is no technological hurdle to build this system. The system is built using widely available technologies such as PHP, MySQL, HTML, CSS. These technologies are easy to implement, update, and maintain. The project is technically feasible with the current tools and skill sets available to the developer.

3.1.2.2 Economic feasibility:

The system does not require extra software and hardware i.e. it uses open-source technologies. So, there is no recurring cost than just the internet connection.

3.1.2.3 Operational feasibility:

This system uses simple technologies to design. So, it is user friendly. The system is practical and easy to operate. Both the admin and users can interact with the system through intuitive user interfaces.

3.1.2.4 Schedule:

Week	Task	Description
Week 1	Requirement Analysis	Collect information about system needs and features.
Week 2	System Design	Prepare ER Diagram, DFD, and database design.
Week 3	Frontend Development	Design login page, dashboard, attendance list.
Week 4	Backend Development	Connect database, create attendance logic, store data.
Week 5	Testing	Check errors, fix bugs, test all modules.
Week 6	Documentation	Prepare project report and finalize system.

Table 3.1: Project Timeline

3.1.3 Methodology Used:

Agile Method:

The Agile methodology is well suited for the Online Student Attendance Management System as it allows the system to be developed incrementally with continuous feedback and improvement. This ensures better adaptability, higher user satisfaction, and efficient project execution. The system is developed in small functional cycles called iterations or sprints. The entire project is divided into small modules. Each module goes through the phases of planning, design, development, testing, and review within a short time frame. After each iteration, a working version of the system is delivered and feedback is collected, which is then used to improve the next iteration.

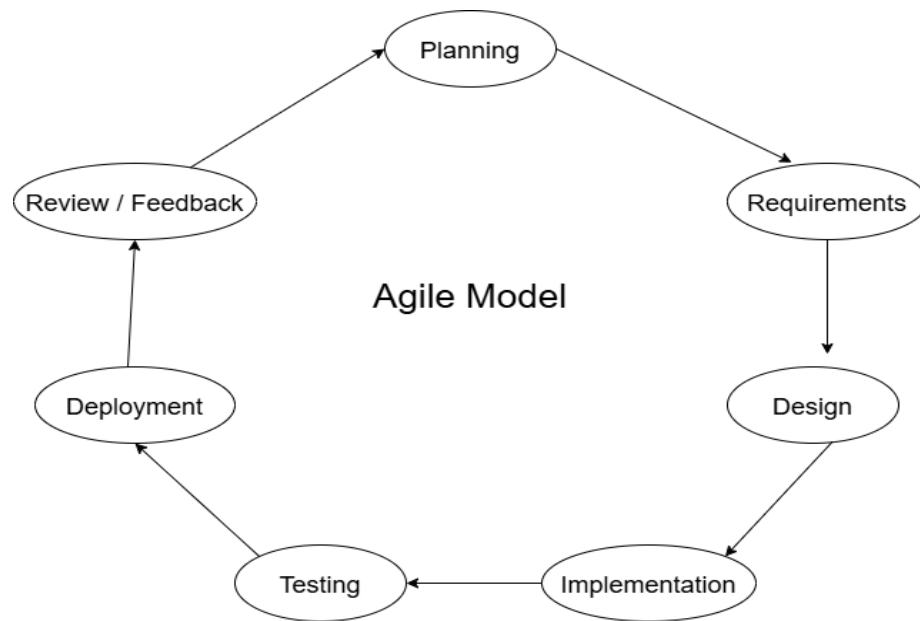


Figure 3.2: Agile Method

Agile Phases Applied to the Attendance System:

- **Requirement Gathering & Planning:** Requirements are collected from stakeholders such as administrators, teachers, and students. These requirements are prioritized and planned for upcoming iterations.
- **Design:** System design is created for the selected features, including database structure, user interfaces, and role-based access.
- **Development:** The selected features (e.g., login system, attendance marking) are developed during a sprint.
- **Testing:** Each developed feature is tested to ensure correctness, security, and performance.
- **Review & Feedback:** The developed module is reviewed by users, and feedback is collected.
- **Deployment & Maintenance:** The working module is deployed, and improvements are made in future iterations.

3.1.4 Data Modelling:

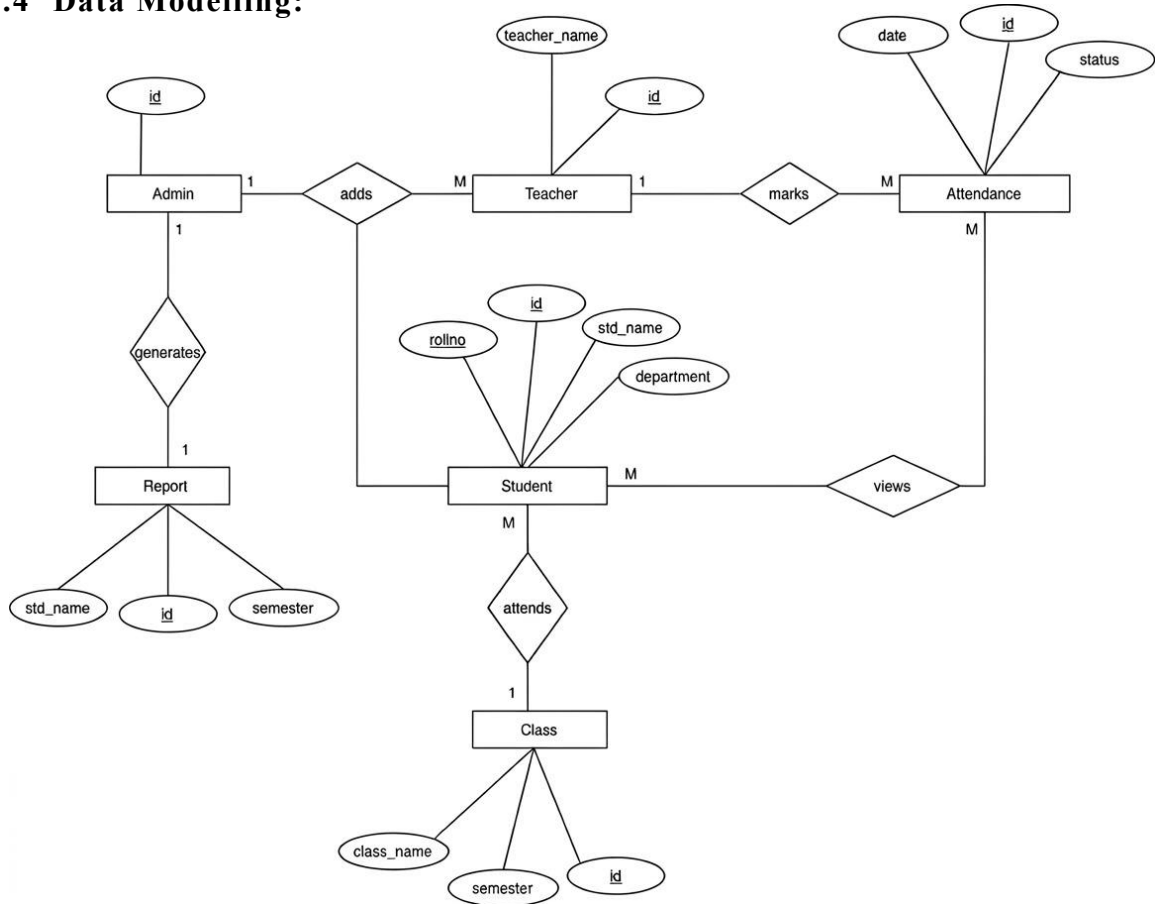


Figure 3.3: ER Diagram

This Entity-Relationship (ER) diagram illustrates online attendance management system centered around four primary entities: Admin, Teacher, Student, and Class. The system defines how these roles interact, such as an Admin who "adds" both Teachers and Students (indicated by a one-to-many relationship), generates attendance report and Teachers who are responsible for "marking" Attendance. Each entity is defined by specific attributes, such as Teacher having a multi-valued "phone" attribute and a primary key "id," while Student is identified by "id" and "rollno." Students have a many-to-one relationship with Class through the "attends" action, implying that multiple students can belong to a single class. Furthermore, there is a many-to-many dynamic regarding attendance; Attendance records (which include attributes like "date" and "status") are marked by Teachers, and Students are then able to "view" these records. This structure ensures a clear hierarchy of data, where administrative oversight, instructional management, and student participation are logically linked.

3.1.5 Process Modelling:

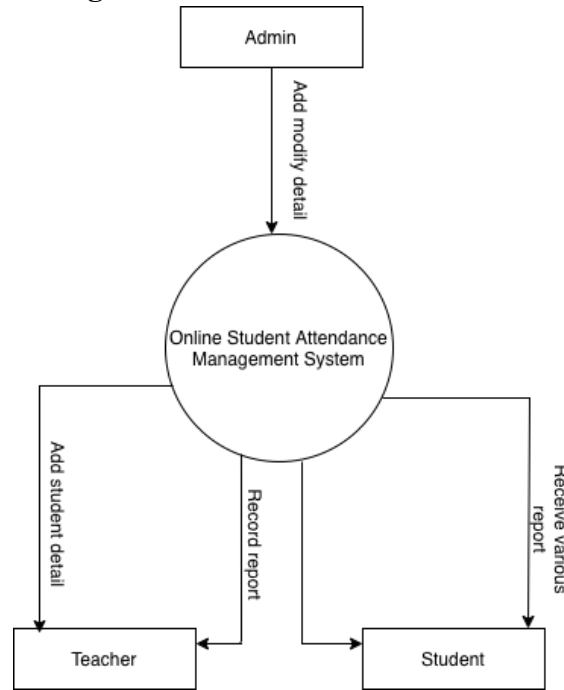


Figure 3.4: Data Flow Diagram Level 0

This Context Diagram (Level 0 DFD) for the Online Student Attendance Management System depicts a centralized digital hub designed to bridge the communication gap between administrative staff, educators, and the student body. At the top of the hierarchy, the admin acts as the primary data architect, feeding the system with "Add/modify detail" inputs that likely encompass the registration of courses, the creation of class schedules, and the management of user credentials. Once this foundational data is processed, the system facilitates a functional loop with the Teacher, who retrieves student details to conduct attendance and subsequently receives "Record reports" to track longitudinal participation and academic compliance. Simultaneously, the system serves the student entity as an information portal, where the output consists of "Receive various reports"—a flow that empowers students to monitor their own attendance percentages and stay informed of any automated warnings or summaries. By positioning the management system as the central process, the diagram illustrates a streamlined workflow where manual record-keeping is replaced by an automated repository, ensuring that data entered by administrators is accurately transformed into actionable insights for teachers and transparent progress tracking for students.

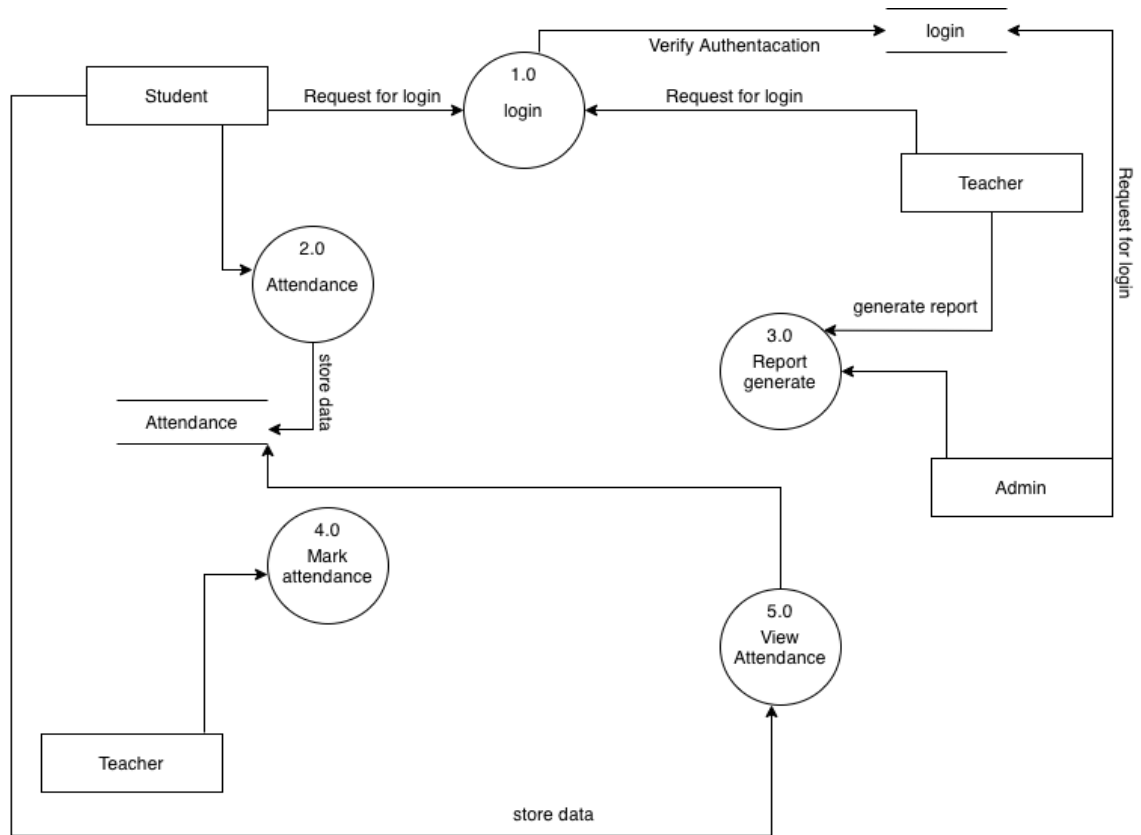


Figure 3.5: Data Flow Diagram Level 1

The Level 1 Data Flow Diagram (DFD) for the Student Attendance Management System provides a granular look at the internal architecture, detailing how raw data is captured, authenticated, and transformed into actionable information through five interconnected processes. The operational cycle begins with Process 1.0 (Login), where Students, Teachers, and Admins initiate a "Request for login"; this process interacts directly with the Login Data Store to "Verify Authentication," ensuring that only authorized users can bypass the system's security perimeter. Once authenticated, the system branches into core functional modules: Process 2.0 (Attendance) and Process 4.0 (Mark Attendance). These processes allow both Students and Teachers to submit daily presence data, which is then routed to the Attendance Data Store—the central repository of the system. This stored data serves as the foundation for the final output stages: Process 5.0 (View Attendance) enables Students to pull their specific records for personal monitoring, while Process 3.0 (Report Generate) allows Teachers and Admins to synthesize the raw attendance logs into comprehensive, high-level summaries.

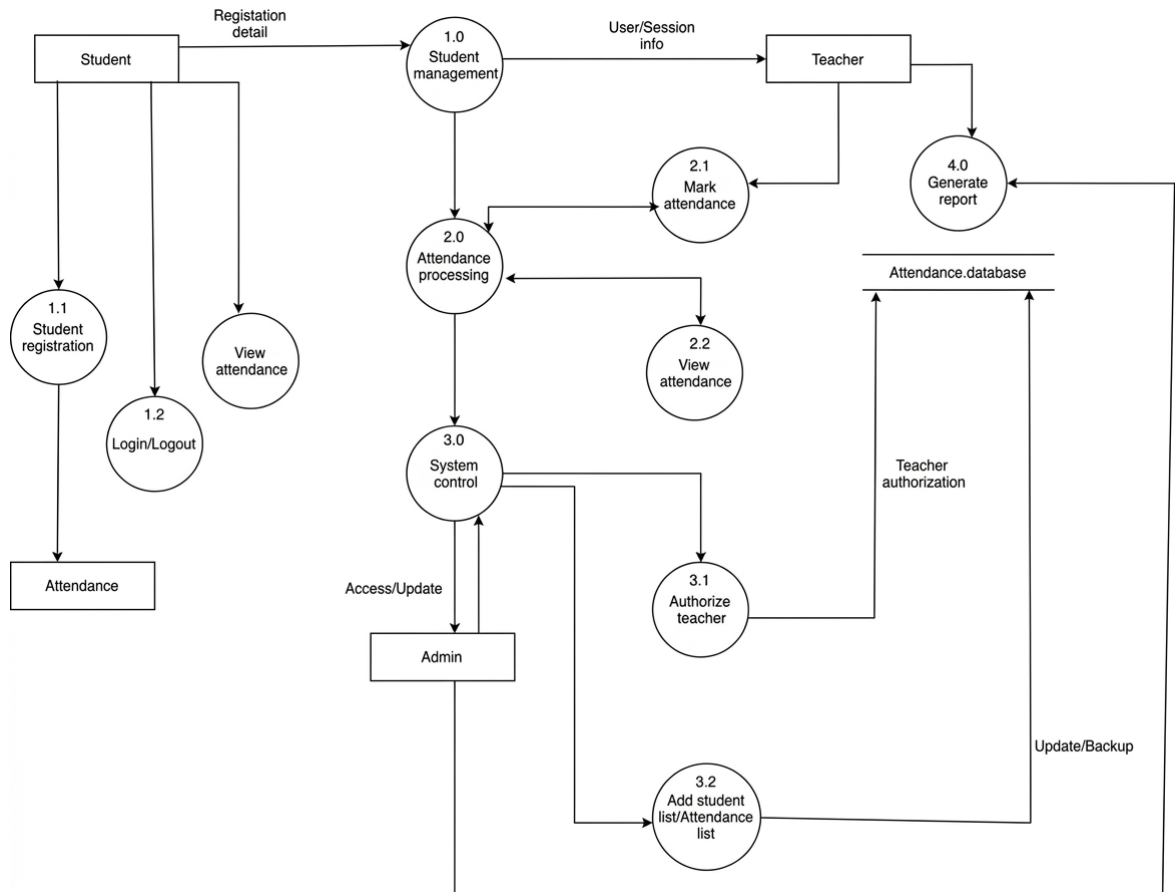


Figure 3.6: Data Flow Diagram level 2

This Data Flow Diagram (DFD level 2) outlines an Online Student Attendance Management System organized into three functional layers. At the entry level, Students register and authenticate via the Student Management (1.0) process, which securely passes session information to the faculty portal. The operational core is handled by Teachers through Attendance Processing (2.0), where they mark daily attendance and view records, while a specialized Generate Report (4.0) module pulls stored data to create high-level summary outputs. The backend is governed by the Admin through the System Control (3.0) hub. This administrative layer manages security by authorizing teacher access and performing critical database backups. All processes ultimately converge at the Attendance Database, which acts as the central repository for all user information and attendance logs. By separating these roles, the system ensures a secure, automated workflow from initial student registration to final report generation.

3.2 System Design:

The system design of Online Student Attendance Management System consists of architectural design, database schema design shown as follows:

3.2.1 Architectural Design:

The system follows a 3-Tier Architecture, which separates the user interface, business logic, and data storage to ensure scalability and security.

- **Presentation Tier (UI):** The web/mobile interface where Teachers, Students, and Admins interact with the system.
- **Application Tier (Logic):** The server-side logic (using language PHP) that processes login authentication and attendance calculations.
- **Data Tier (Database):** The MySQL database that stores the entities defined in ER diagram.

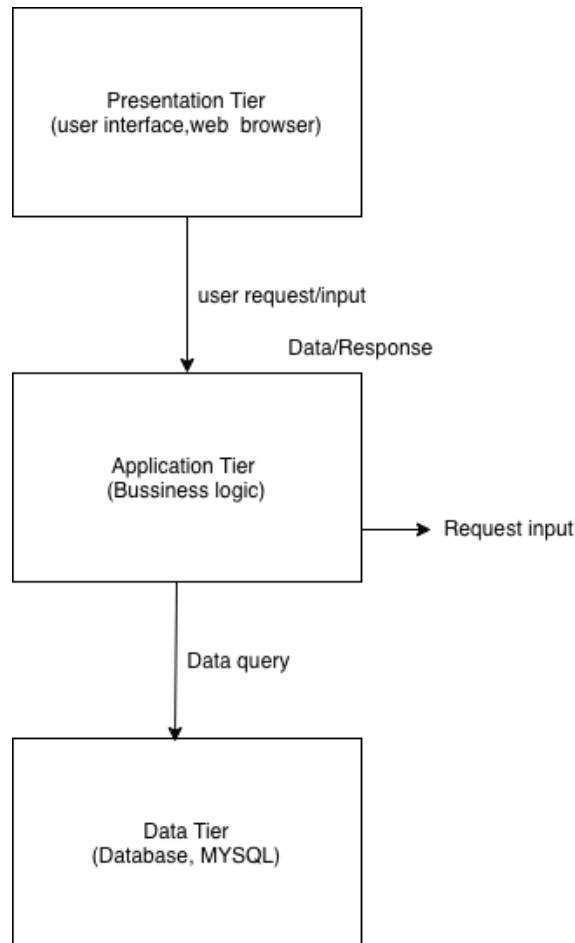


Figure 3.7: Architectural Design

3.2.2 Database Schema Design:

This section converts the ER Diagram into relational tables. Each "Entity" is represented as a table, and its "Attributes" are mapped to specific columns with defined data types and constraints.

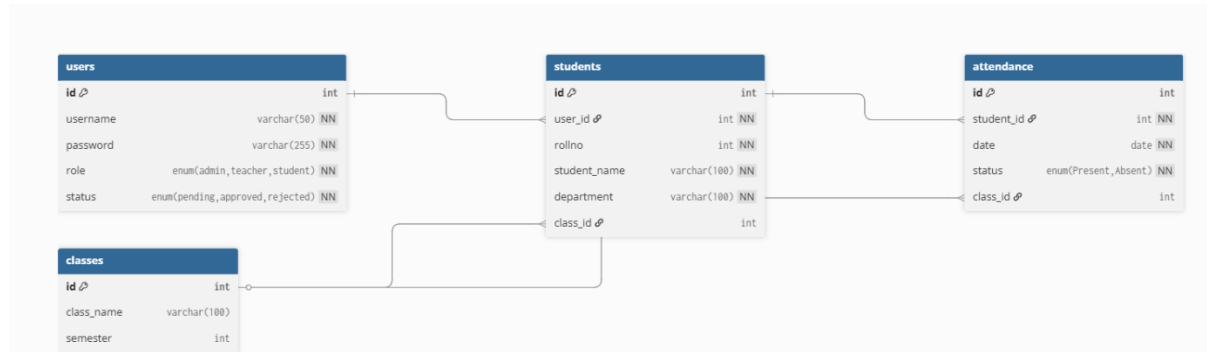


Figure 3.8: Database Schema

3.2.3 Physical DFD:

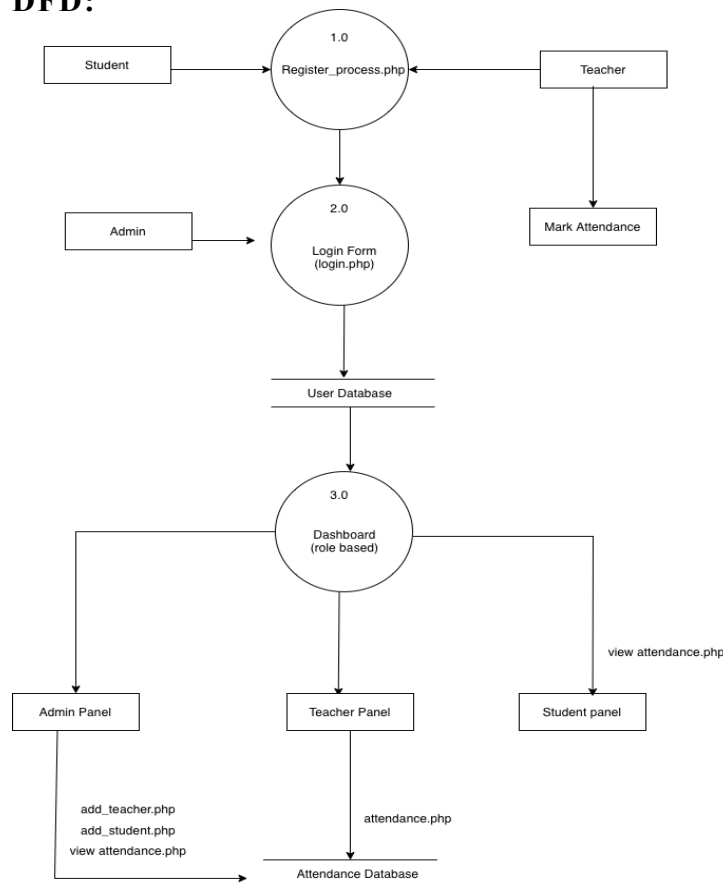


Figure 3.9: Physical DFD

CHAPTER IV: IMPLEMENTATION AND TESTING

4.1 Implementation:

Implementation is the process of putting a plan, design, or system into operation. It involves developing, configuring, and deploying the components of a system so that it functions as intended to achieve its objectives. The system is implemented as a structured web application where users can log in securely according to their roles. Attendance is recorded, and stored in a centralized database, ensuring accuracy and easy retrieval of records. The system provides a user-friendly interface for managing attendance while maintaining data security through controlled access and validation. It is developed and tested for smooth functionality in an academic environment.

4.1.1 Tools Used:

4.1.1.1 Programming Languages:

Frontend:

- **HTML5:** HTML5 was used to design the structure of the web pages, such as login forms, dashboards, and attendance tables.
- **CSS3:** CSS3 was used to style the web pages, making the interface user-friendly and visually organized.

Backend:

- **PHP:** PHP was used for server-side scripting to process forms, handle login authentication, and connect the frontend with the database.
- **Database: MySQL** was used to store all system data including students' details, teachers' details and attendance logs.

4.1.1.2 Tools / Software:

- **XAMPP:** XAMPP was used to run the Apache server and MySQL database locally for testing and deployment.
- **VS Code:** VS Code was used to write and organize all code files.
- **Browser:** Chrome was used to test and interact with the web application.
- **phpMyAdmin:** PhpMyAdmin was used to create, manage, and test the database.

4.1.2 Implementation Details of Modules (Description of procedures/functions):

- **Login Module:**

This module provides secure access to the system. Teachers, students, and admin log in using their username and password. The system verifies the credentials from the database and allows access based on user role.

- **Admin Module:**

The admin module manages the overall system. The admin can add teachers (give access to teachers), add students, and maintain user records/view attendance. This module ensures proper control and management of the attendance system.

- **Teacher Module:**

The teacher module allows teachers to manage attendance activities. Teachers can mark daily attendance, view students' attendance lists and check previously recorded attendance data.

- **Student Module:**

The student module allows students to log in/register and view their attendance details.

- **Attendance Management Module:**

This module handles the main functionality of the system. It records attendance data such as date, course, and student status (present/absent) and stores it in the database securely.

- **Database Module:**

The database module stores all information related to admin, teacher, student, and attendance records. It ensures data integrity, accuracy, and fast retrieval of information.

4.2 Testing:

The Online Student Attendance Management System was thoroughly tested to ensure its functionality, accuracy, and usability. Different types of testing were performed, including unit testing to check individual modules, and system testing to evaluate overall performance.

4.1.3 Test Cases for Unit Testing:

Table 4.1: Test Case 001 – Student Login:

Pre-conditions: Student has valid login credentials

Dependencies: Registration Module, Authentication module

S.N.	Test Steps	Input	Expected Result	Actual Result	Status
1	Open login page	System URL	Login page is displayed	Login page displayed successfully	Pass
2	Enter valid username and password	student1 / 1234	Student dashboard is displayed	Dashboard displayed successfully	Pass
3	Enter invalid username	wrong1 / 1234	Login is denied	Invalid credentials message displayed	Pass
4	Enter valid username and invalid password	student1 / wrong pass	Login is denied	Invalid credentials message displayed	Pass

Table 4.2: Test Case 002 – Mark Attendance:

Pre-conditions: Teacher is logged into the system

Dependencies: Login Module

S.N.	Test Steps	Input	Expected Result	Actual Result	Status
1	Open attendance page	Select date	Attendance page is displayed	Attendance page displayed successfully	Pass
2	Mark students present/absent	Select students & submit	Attendance is saved	Attendance recorded successfully	Pass

Table 4.3: Test Case 003 – View Attendance:

Pre-conditions: Attendance data is available

Dependencies: Attendance Module

S.N.	Test Steps	Input	Expected Result	Actual Result	Status
1	Open view attendance section	Select date	Attendance report is displayed	Report displayed correctly	Pass

4.1.4 Test Cases for System Testing:

Table 4.4: Test Case 004 – System Integration & Navigation:

Pre-conditions: User (student/teacher) logged in

Dependencies: Login, Attendance, Viewing modules

S.N.	Test Steps	Input	Expected Result	Actual Result	Status
1	Navigate from dashboard to attendance	Click Mark Attendance link	Attendance page opens	Page displayed successfully	Pass
2	Mark attendance and save	Select students, submit	Attendance saved	Attendance recorded successfully	Pass
3	Navigate to view attendance	Click View Attendance link	Reports page opens	Page displayed successfully	Pass
4	View attendance	Select date	Correct report displayed	Report displayed correctly	Pass
5	Logout	Click Logout	User redirected to login page	Logout successful	Pass

CHAPTER V: CONCLUSION AND FUTURE RECOMMENDATIONS

5.1 Conclusion:

The Online Student Attendance Management System (OSAMS) successfully demonstrates a basic and focused approach to managing student attendance. Since this project is developed for academic purposes, the functions implemented are few and specific, including user registration, login/logout, marking attendance, and viewing attendance records. These core features provide a clear understanding of how attendance management can be automated, while keeping the system simple and practical for learning purposes. The project helped in gaining hands-on experience in software development, module integration, and user interface design, while emphasizing data accuracy and secure access. Overall, the project achieves its intended objectives within a limited scope and serves as a functional example of a student attendance management system designed for academic learning. This system achieves its intended academic objectives by presenting a functional, secure, and organized solution for attendance management. It demonstrates practical implementation of theoretical concepts and strengthens technical skills in system design and development. Despite its simplicity, the project highlights the importance of automation in educational institutions and provides a scalable base for future enhancements. Overall, OSAMS stands as a successful academic project that combines learning, functionality, and practical application in a meaningful way.

5.2 Lesson Learnt / Outcome:

The development of the Online Student Attendance Management System (OSAMS) provided an important learning experience in the field of software development. Through this project, we gained a deeper understanding of the entire software development life cycle, including requirements gathering, system design, implementation, testing, and deployment. We learned how to design a system with different modules working together efficiently, such as registration, login, attendance marking, report generation, and user management. The project also emphasized the importance of data security, accuracy, and integrity, ensuring that attendance records are reliable and protected. In addition, we improved our skills in problem-solving, debugging, and user interface design, making the system both functional and user-

friendly. We also realized the value of documentation, planning, and time management in completing a project successfully. Overall, the project resulted in a fully functional and practical attendance management system, while providing valuable insights and hands-on experience.

Outcome of the Project:

- Users (Teachers/Students) can register and have their accounts added to the system.
- Users (Admin/Teachers/Students) can log in and log out securely.
- Admin provides authorization/ access to the users(teachers).
- Teachers can mark attendance for students accurately.
- Students can view their attendance easily.
- The system ensures data integrity and reduces manual attendance work.
- Provides a functional, user-friendly system suitable for classroom attendance management.

5.3 Future Recommendation:

Although the current Online Student Attendance Management System successfully fulfills its basic purpose of registering users, allowing login/logout, marking attendance, and viewing attendance records, there are several areas where the system can be further enhanced and extended to improve functionality, usability, and overall efficiency for both students and teachers.

The current version of the Online Student Attendance Management System includes only the basic and essential features. In the future, the system can be enhanced in the following ways:

- Add a feature to generate detailed attendance reports for students, teachers, and administrators.
- Implement notifications via email or SMS for low attendance or missed classes.
- Develop a mobile version of the system for easy access on smartphones and tablets.
- Integrate automatic attendance tracking using class schedules, RFID, or QR codes.
- Add data analytics and charts to show attendance trends and statistics.
- Expand the system to handle multiple classes and subjects, making it more flexible for larger schools.

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APPENDICES

Login

Login

Don't have an account?

Register

Teacher
Dashboard

Mark
Attendance

View
Attendance

Attendance
Report

Logout

Welcome Teacher

Use the navigation above to manage student attendance or view your records.

Mark Student Attendance

Select Date: 02/18/2026

Select Class: BCA(Sem: 1)

Load Student List

Roll No	Student Name	Mark Status
1	Andrew Davilla	☒ Present ☐ Absent
2	Bella Smith	☐ Present ☒ Absent

Submit Attendance for 2026-02-18

View Attendance Records

Class:

BCA (1)

Date:

2026-02-19

View Attendance

Attendance for: 2026-02-19

Roll No	Student Name	Status
1	Andrew Davilla	Present
2	Bella Smith	Absent

BCA Semester 1 Attendance Report

Sem 1

Sem 2

Sem 3

Sem 4

Month:

March



Find Student:

Andrew Davilla

Generate Report

[Reset All](#)

Student Name	Total Days	Present	Absent	Attendance %
Andrew Davilla	6	2	4	33.33%

Download PDF Report

BCA Semester 2 Attendance Report

Sem 1

Sem 2

Sem 3

Sem 4

Month:

February



Find Student:

Simrika Basnet

Generate Report

[Reset All](#)

Student Name	Total Days	Present	Absent	Attendance %
Simrika Basnet	1	0	1	0.00%

Download PDF Report